

Select & Distinct Statements

→ SELECT * from students; → whole table.

→ SELECT branch from students;

↓
Only the branch column.

→ SELECT DISTINCT branch from students;

↓
This selects only specific data from the branch.

→ SELECT * from employees WHERE

emp-id=4;

↓
This will retrieve the complete data ~~all column~~ about the 4th employee.

→ SELECT * from employees WHERE empid BETWEEN 1 and 3.

↓
Retrieves the data b/wn (∵ 1 & 3 inclusive)
where $emp-id \leq 3$.

Operators

* = (Equal)

* IN

* < > (Not equal)

* BETWEEN

* LIKE

ORDERBY

* This used to sort the result-set in ascending or descending order.

```
SELECT * Score  
FROM ASSESSMENT-SCORES  
WHERE Score >= 60  
ORDERBY Score ASC;
```

* ORDER BY sorts through the ASCII value.

* By default the order is ASC.

Update & Delete

* Modifies the existing records in a table.

Syntax →

```
UPDATE table-name  
SET Column 1 = value 1, etc...  
WHERE condition( );
```

Ex:- UPDATE employees SET salary = 150000
dept = 'Management' WHERE emp-id = 4;

Delete

* DELETE FROM employees WHERE
dept = 'Logistics';

TOP & LIMIT

SELECT column-name
FROM table-name
WHERE condition
ORDER BY column-names
LIMIT number;

* SELECT * FROM employees LIMIT 5;
↓
retrieves the first 5 data
entries.

* SELECT * FROM employees ORDER BY
name LIMIT 5;
↓
retrieves the first 5 data
entries alphabetically.

Min & Max

SELECT ^{MAX} / MIN (column-name)
FROM table-name
WHERE condition.

SELECT 2nd or nth highest data entry

SELECT MAX(salary) FROM employee WHERE
salary < (SELECT MAX(salary) FROM
employee);

* This retrieves the 2nd highest salary in the database.

Offset & Row count

→

1, 2
↓ ↓
offset row count

1
2
3
4
5
6
7
...

Ex:- 3, 4

First 3 values will not be considered

Row-count = 4

Syntax:-

SELECT column FROM table ORDER BY column
LIMIT offset, row-count;

Ex:- SELECT salary FROM employees ORDER BY
salary LIMIT 3, 4;

COUNT, AVG, SUM Avg/sum.

SELECT COUNT(column-name)
FROM table-name
WHERE condition;

} Syntax

* SELECT SUM(salary) FROM employees

⇓
retrieves sum of all
people salaries.

* SELECT SUM(salary) FROM employees
WHERE branch-id = 3;

Like & Wild Cards

* ⇒ Represents zero or more characters

? ⇒ Represents a single character

[] ⇒ Represents any single character
within the brackets

! ⇒ Represents any character not in the
brackets

- ⇒ Represents a range of characters.

⇒ Represents any single numeric character

% => Represents zero or more characters

_ => Represents a single character

[] => Represents any single within brackets

^ => Represents any character not in the brackets

- => Represents a range of characters

Ex:-

```
SELECT * FROM employee WHERE name  
LIKE 'a%o%'
```

Irrespective of how many letters that word consists it should start from 'a'.

'%u%' => The word should contain 'u' mandatorily.

'%ch%' => Should contain 'ch' in the word.

'_i%' => Second character should be 'i'.

'_ch%' => Should be one character before 'ch'.

'mi_%' => First two characters should be 'mi' & the next two characters can be anything from there many characters can take place.

'm%t' => Starts with 'm' & ends with 't'.

* NOT LIKE '%a' $\Rightarrow$ Name should not end with a
* '[ahm]%' $\Rightarrow$ should start with either d, h or m.

* '%[y]%' $\Rightarrow$ ~~it should be in middle~~
should contain "y" or "l".

* '%[hm][FS]%' $\Rightarrow$ $\xrightarrow{\text{h (or) m}}$ $\xrightarrow{\text{F (or) S}}$

* '[a-d]%' $\Rightarrow$ Any character b/w a & d is acceptable at the start

* '%[l-m]%' $\Rightarrow$ ~~any~~ any character b/w l & m is acceptable in the word.

NOT LIKE

* '[!am]%' $\Rightarrow$ Name should not start with "a" or "m".

* '[!am]%' $\Rightarrow$ Does not return the characters in the brackets

$\rightarrow$ Floor('b') $\Rightarrow$ 0 (Floor of a string gives zero in my sql).

$\rightarrow$ Concat('sql', 'q'); $\Rightarrow$ This will return sql.q

$\rightarrow$ TO - NUMBER('5') $\Rightarrow$ Converts a character value into a numeric datatype.